

CHE 301

1. Course number and name

CHE 301. Chemical Engineering Thermodynamics

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Michael Caracotsios

4. Textbook, title, author, publisher, year

Fundamentals of Chemical Engineering Thermodynamics, Themis Matsoukas, Prentice Hall, 2013.

a. other supplemental materials

- Chemical Engineering Thermodynamics Lecture Notes, will be distributed to all students free of charge via the UIC Blackboard, Michael Caracotsios, 2020
- J. M. Smith, H. C. Van Ness and M. M. Abbott, Introduction to Chemical Engineering Thermodynamics, 7th Edition, McGraw-Hill, 2005.
- S. I. Sandler, Chemical and Engineering Thermodynamics, 3rd Edition, John Wiley & Sons, Inc., 1999.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Review of first and second laws of thermodynamics; Equations of state; Multi-component systems and phase equilibria; Chemical reaction thermodynamics and reaction equilibria.

b. Prerequisites or co-requisites

CHE 201 and CHE 205.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

Understand the basic concepts of Internal Energy, Gibbs Energy, Entropy and Enthalpy.

Apply control volume analysis to a variety of situations related the phase and chemical equilibrium and solve problems utilizing computer software.

Understand the theory of thermodynamics of ideal and non-ideal solutions

Understand the applicability and use of various equations of state in describing the thermodynamic state of pure fluids and mixtures thereof.

Determine equilibrium conditions between phases using equations of state and liquid activity coefficients.

Determine equilibrium conditions between reactants and products using equilibrium constants and Gibbs energy minimization concepts.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(3) an ability to communicate effectively with a range of audiences.		X		

7. Brief list of topics to be covered

REVIEW AND BACKGROUND: Introduction to Thermodynamics in Chemical Engineering, Elements of Differential and Integral Calculus, Numerical Methods for Algebraic and Differential Equations, Mass and Energy Balances in the Context of Thermodynamics, Tutorials on Programming with VBA and Athena Visual Studio

THERMODYNAMIC PROPERTIES OF SINGLE-COMPONENT FLUIDS: PVT Behavior of Single-component Fluids, Gibbs Phase Rule for Single-component Fluids, The Ideal Gas Behavior, Cubic and Virial Equations of State, Fugacity and Fugacity Coefficients from Equations of State, Thermodynamic Properties of Single-Phase and Two-Phase Systems

THERMODYNAMIC PROPERTIES OF MULTI-COMPONENT MIXTURES: Partial Molar Properties Excess and Residual Thermal Properties, the Gibbs Phase Rule for Multi-component Mixtures, PVT Behavior of Multi-component Mixtures, Cubic Equations of State and Mixing Rules, Fugacity and Fugacity Coefficients from Equations of State, Liquid Activity Coefficients for Non-ideal Solutions, Thermodynamic Properties of Single-Phase and Two-Phase Systems

MULTICOMPONENT PHASE EQUILIBRIUM: Criteria for Phase Equilibrium, Vapor-Liquid Equilibrium: The Fundamental Problem, Flash Calculations: PT, PH, PV, TH, TV and HV Calculations, VLE Data Analysis: Activity Coefficients from Experimental Data, Liquid-Liquid Equilibrium: The Fundamental Problem

CHEMICAL REACTION EQUILIBRIUM: Criteria for Equilibrium in Chemical Reactions, Gibbs Free Energy and Equilibrium Constants, Evaluation of Equilibrium Constants from Thermodynamic Data, Calculation of Equilibrium Conversion for Single and Multi-reaction Systems